ForecastIQ: A Modern Multi-Horizon Time-Series Forecasting and Benchmarking Platform

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Abstract

This report details the implementation of ForecastIQ, a SOTA sequence forecasting framework comparing Patch-based Time-Series Transformers (PatchTST), univariate residual decomposition networks (N-BEATS), and Temporal Fusion Transformers (TFT). Utilizing channel-independence, multi-frequency simulated demand datasets, and robust MinMax scaling pipelines, the platform evaluates multi-horizon predictions (24 steps out based on 96 steps of history) using MAE, RMSE, and sMAPE. A FastAPI web prediction engine and Streamlit comparative dashboard with Plotly residual charts are developed.

I. Introduction

Time-series forecasting is essential across demand planning, logistics, and resource allocation. While traditional methods (like LSTMs) struggle to preserve long-term temporal dependencies, modern transformer architectures address these limitations. ForecastIQ implements and benchmarks three SOTA architectures: PatchTST, N-BEATS, and a custom Gated Gated TFT.

II. Forecasting Architectures

A. Model Specifications

- **PatchTST**: Implements channel-independence (flattening features to parallel subseries) and groups sequential steps into overlapping patches. This reduces transformer sequence complexity from quadratic to linear while preserving local patterns.
- **N-BEATS**: A deep stack of fully-connected blocks utilizing forward and backward residual links to decompose sequence trends and seasonality components.
- **Temporal Fusion Transformer (TFT)**: Integrates multi-horizon prediction, self-attention, static covariate encoders, and gated linear unit (GLU) nodes to filter out noise.

III. Data and Processing Pipeline

The data pipeline implements a rolling-window formatter. Input arrays of shape $[\text{Batch}, \text{Seq\_Len}, \text{Features}]$ are normalized relative to running scale factors. The dataset split isolates the final 20% segment for validation and backtest simulations, ensuring no data leakage during model training.

IV. Metric Formulation

Three key metrics are calculated to evaluate accuracy:

$$\text{MAE} = \frac{1}{N}\sum |y_i - \hat{y}_i| \tag{1}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum (y_i - \hat{y}_i)^2} \tag{2}$$

$$\text{sMAPE} = \frac{100\%}{N}\sum \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \tag{3}$$

V. Conclusion

ForecastIQ presents a standardized benchmarking environment for time-series forecasting. By deploying SOTA architectures (PatchTST, N-BEATS, and TFT) side-by-side with strict validation and REST API pipelines, the platform offers a production-ready solution for enterprise sequence prediction.